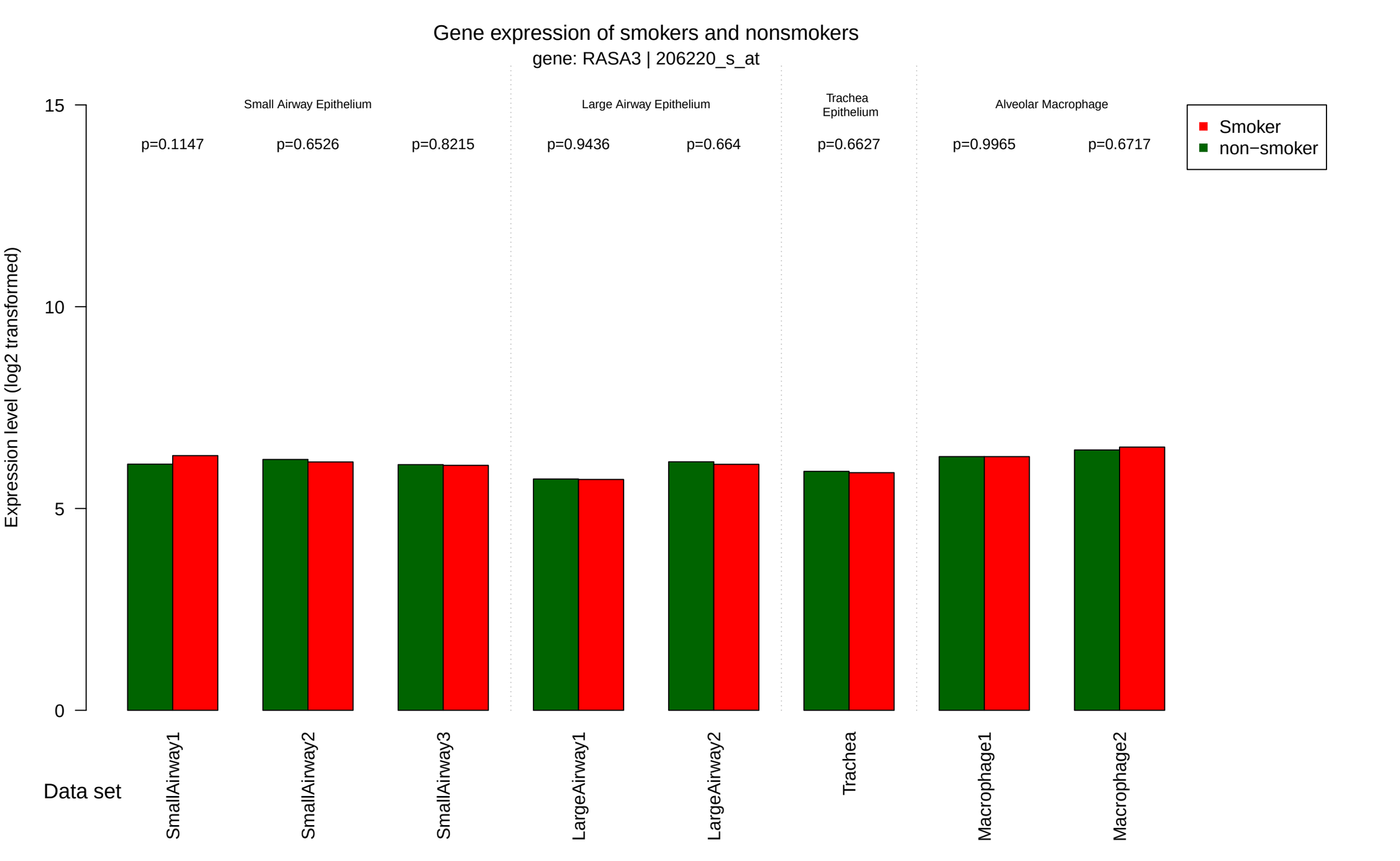
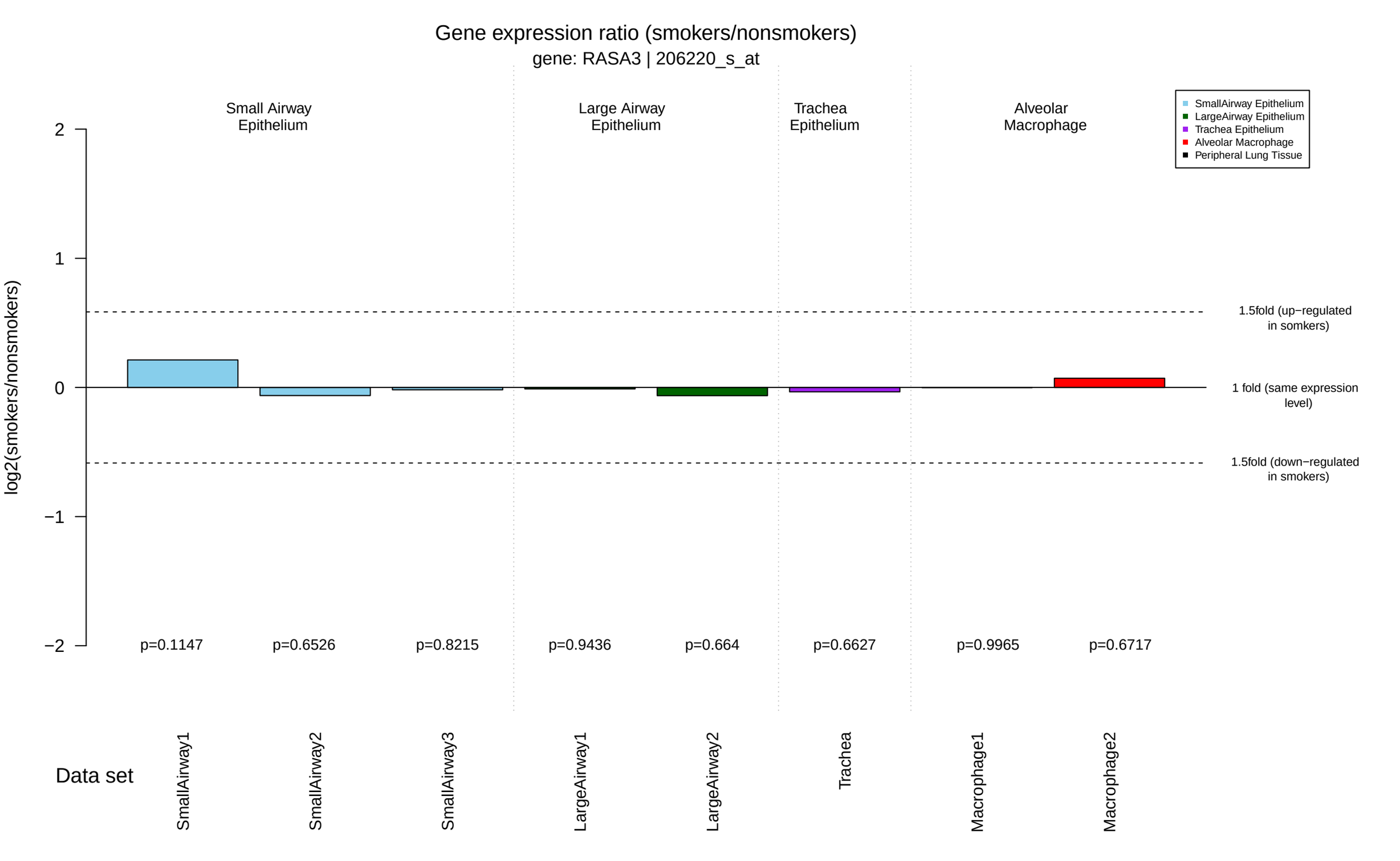
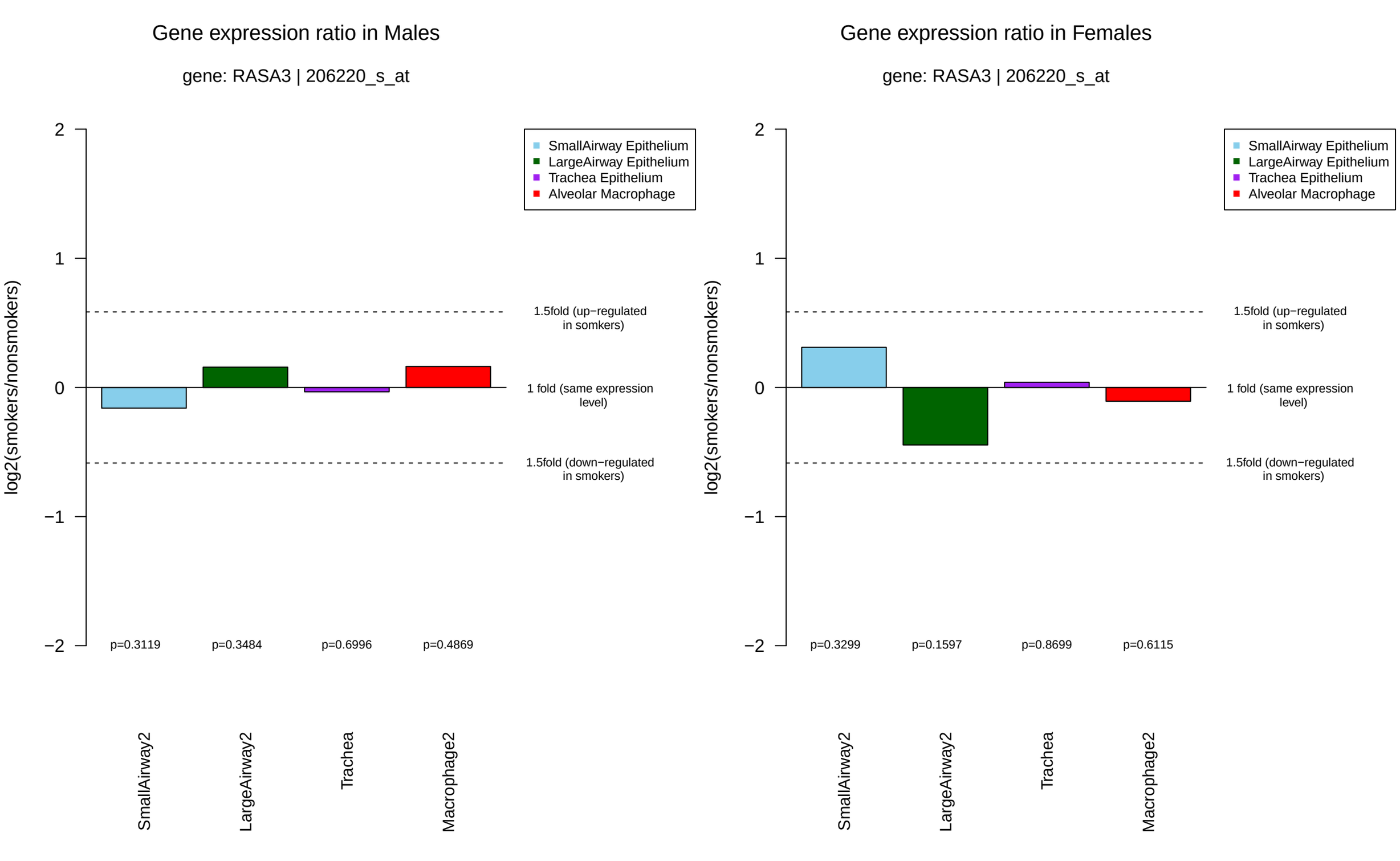
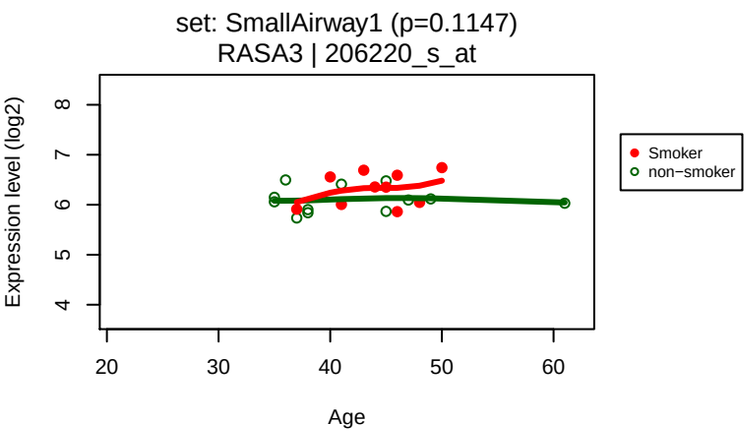
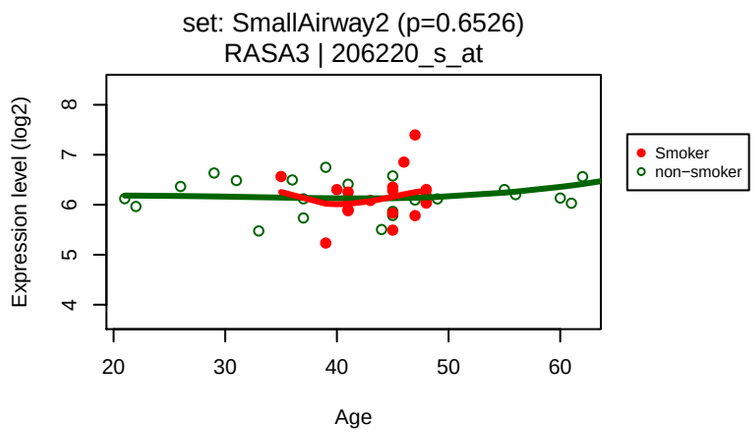
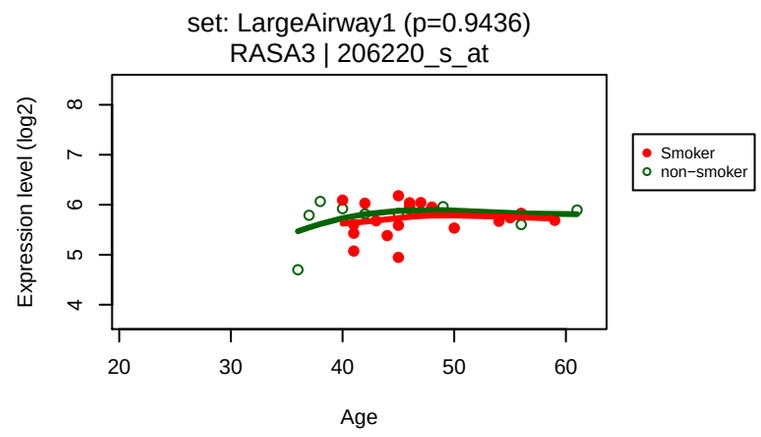

Gene expression of smokers and nonsmokers
gene: RASA3 | 206220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1147
p=0.6526
p=0.8215
p=0.9436
p=0.664
p=0.6627
p=0.9965
p=0.6717
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

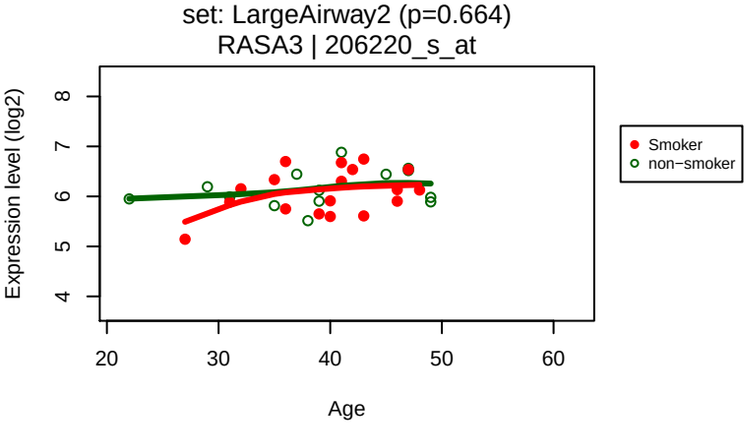

Gene expression ratio (smokers/nonsmokers)
gene: RASA3 | 206220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1147
p=0.6526
p=0.8215
p=0.9436
p=0.664
p=0.6627
p=0.9965
p=0.6717
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

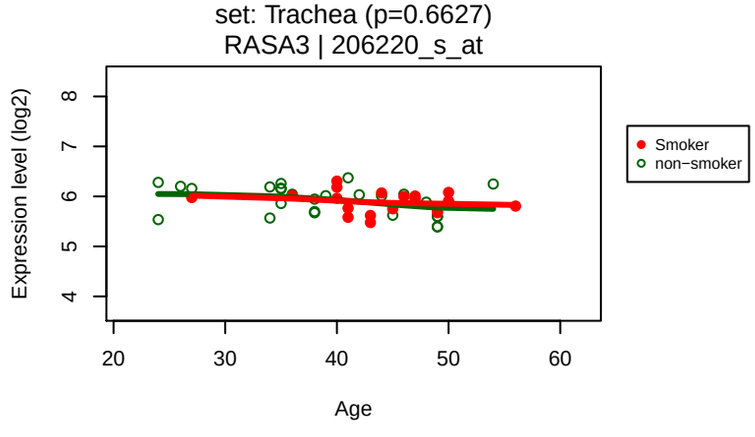

Gene expression ratio in Males
gene: RASA3 | 206220_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3119
p=0.3484
p=0.6996
p=0.4869
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASA3 | 206220_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3299
p=0.1597
p=0.8699
p=0.6115
SmallAirway2
LargeAirway2
Trachea
Macrophage2

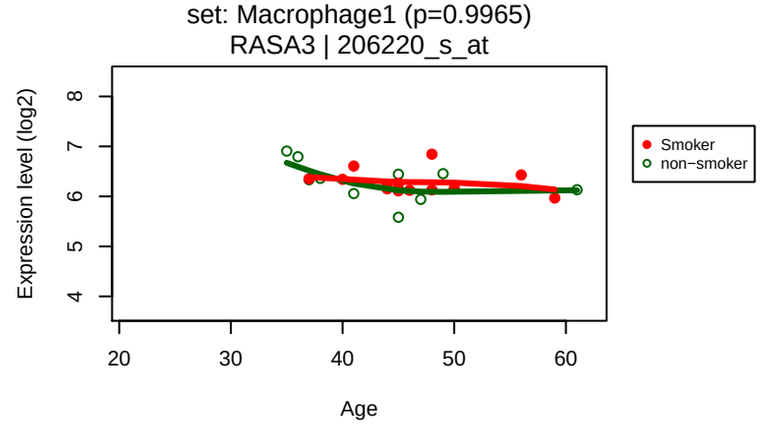

set: SmallAirway1 (p=0.1147)
RASA3 | 206220_s_at
Expression level (log2)
Age
Smoker
non−smoker

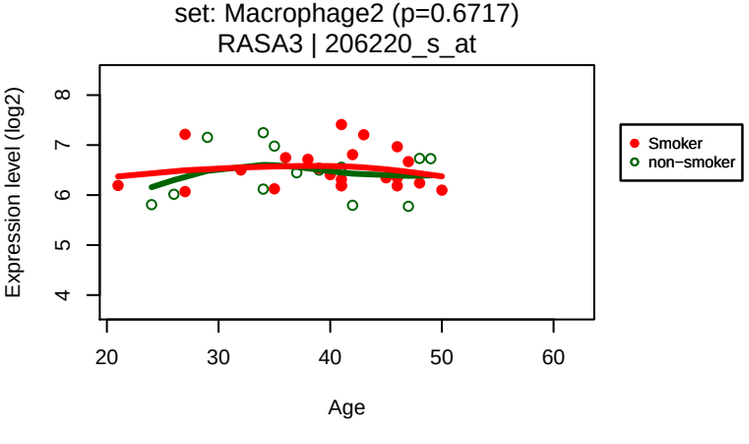

set: SmallAirway2 (p=0.6526)
RASA3 | 206220_s_at
Expression level (log2)
Age
Smoker
non−smoker

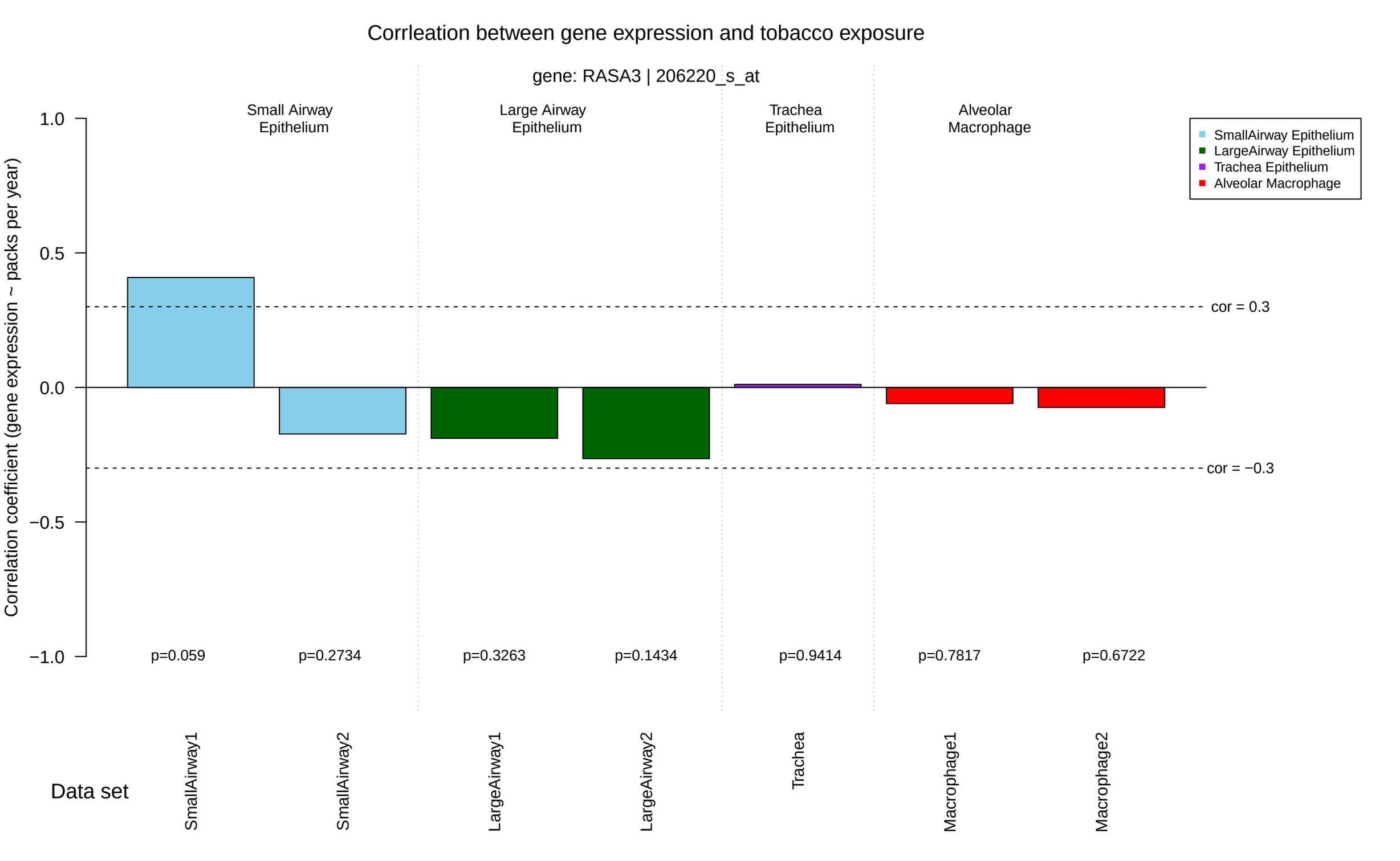

set: LargeAirway1 (p=0.9436)
RASA3 | 206220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.664)
RASA3 | 206220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6627)
RASA3 | 206220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9965)
RASA3 | 206220_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6717)
RASA3 | 206220_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASA3 | 206220_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.059
p=0.2734
p=0.3263
p=0.1434
p=0.9414
p=0.7817
p=0.6722
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2